



Lab 02: IAM Custom Roles

Cloud Security Fundamentals | GSP190

Overview

Created, updated, disabled, deleted, and restored custom IAM roles using both YAML files and CLI flags. Custom roles enforce the principle of least privilege by bundling only the permissions needed for a specific job function.

Objectives

- View available permissions for a resource
- Get role metadata for predefined roles
- View grantable roles on resources
- Create custom roles using YAML and CLI flags
- List, update, disable, delete, and restore custom roles

Task 1 - View Available Permissions

```
gcloud iam list-testable-permissions  
//cloudresourcemanager.googleapis.com/projects/$DEVSHHELL_PROJECT_ID
```

```
CLOUD SHELL
Terminal (qwiklabs-gcp-01-db1c6195cda8) x + v
title: Update Datastores
---
apiDisabled: true
name: vmwareengine.dnsBindPermission.get
stage: GA
title: Get DnsBindPermission
---
apiDisabled: true
name: vmwareengine.dnsBindPermission.grant
stage: GA
title: Grant DnsBindPermission
---
apiDisabled: true
name: vmwareengine.dnsBindPermission.revoke
stage: GA
title: Revoke DnsBindPermission
---
apiDisabled: true
name: vmwareengine.dnsForwarding.get
stage: GA
title: Get DnsForwarding
---
apiDisabled: true
name: vmwareengine.dnsForwarding.update
stage: GA
title: Update DnsForwarding
---
apiDisabled: true
name: vmwareengine.externalAccessRules.create
stage: GA
title: Create ExternalAccessRule
---
apiDisabled: true
name: vmwareengine.externalAccessRules.delete
stage: GA
title: Delete ExternalAccessRule
---
apiDisabled: true
name: vmwareengine.externalAccessRules.get
stage: GA
title: Get ExternalAccessRule
---
apiDisabled: true
name: vmwareengine.externalAccessRules.list
stage: GA
title: List ExternalAccessRules
---
apiDisabled: true
name: vmwareengine.externalAccessRules.update
stage: GA
title: Update ExternalAccessRule
---
apiDisabled: true
```

Task 2 - Get Role Metadata

```
gcloud iam roles describe roles/viewer
```

```

- workflows.locations.list
- workflows.operations.get
- workflows.operations.list
- workflows.stepEntries.get
- workflows.stepEntries.list
- workflows.workflows.get
- workflows.workflows.list
- workflows.workflows.listEffectiveTags
- workflows.workflows.listRevision
- workflows.workflows.listTagBindings
- workloadcertificate.locations.get
- workloadcertificate.locations.list
- workloadcertificate.operations.get
- workloadcertificate.operations.list
- workloadcertificate.workloadCertificateFeature.get
- workloadcertificate.workloadRegistrations.get
- workloadcertificate.workloadRegistrations.list
- workloadmanager.actuations.get
- workloadmanager.actuations.list
- workloadmanager.deployments.get
- workloadmanager.deployments.list
- workloadmanager.discoveredprofiles.get
- workloadmanager.discoveredprofiles.getHealth
- workloadmanager.discoveredprofiles.list
- workloadmanager.evaluations.get
- workloadmanager.evaluations.list
- workloadmanager.executions.get
- workloadmanager.executions.list
- workloadmanager.insights.export
- workloadmanager.insights.listSapSystems
- workloadmanager.locations.get
- workloadmanager.locations.list
- workloadmanager.operations.get
- workloadmanager.operations.list
- workloadmanager.results.list
- workloadmanager.rules.list
- workspace.marketplace.appConfiguration.view
- workstations.operations.list
- workstations.workstationClusters.get
- workstations.workstationClusters.list
- workstations.workstationClusters.listEffectiveTags
- workstations.workstationClusters.listTagBindings
- workstations.workstationConfigs.get
- workstations.workstationConfigs.getIamPolicy
- workstations.workstationConfigs.list
- workstations.workstations.get
- workstations.workstations.getIamPolicy
- workstations.workstations.list
name: roles/viewer
stage: GA
title: Viewer
student_01_55c96268f1c6@cloudshell:~ (qwiklabs-gcp-01-db1c6195oda8) $

```

Task 3 - View Grantable Roles

```

gcloud iam list-grantable-roles
//cloudresourcemanager.googleapis.com/projects/$DEVSHHELL_PROJECT_ID

```

Task 4 - Create Custom Roles

Created two custom roles: editor role via YAML file and viewer role via CLI flags.

YAML file method (role-definition.yaml):

```

title: "Role Editor"
description: "Edit access for App Versions"
stage: "ALPHA"
includedPermissions:
- appengine.versions.create
- appengine.versions.delete

```

```

gcloud iam roles create editor --project $DEVSHHELL_PROJECT_ID --file role-
definition.yaml

```

CLI flags method:

```
gcloud iam roles create viewer --project $DEVSHHELL_PROJECT_ID --title "Role Viewer"
--description "Custom role description." --permissions
compute.instances.get,compute.instances.list --stage ALPHA
```

```
description: Grants the Workstations Service Account access to manage resources in
  consumer project.
name: roles/workstations.serviceAgent
title: Workstations Service Agent
student_01_55c96268f1c6@cloudshell:~ (qwiklabs-gcp-01-dblc6195cda8)$ nano role-definition.yaml
student_01_55c96268f1c6@cloudshell:~ (qwiklabs-gcp-01-dblc6195cda8)$ gcloud iam roles create editor --project $DEVSHHELL_PROJECT_ID --file role-definition.yaml
Created role [editor].
description: Edit access for App Versions
etag: BwZTKFthyWI=
includedPermissions:
- appengine.versions.create
- appengine.versions.delete
name: projects/qwiklabs-gcp-01-dblc6195cda8/roles/editor
stage: ALPHA
title: Role Editor
student_01_55c96268f1c6@cloudshell:~ (qwiklabs-gcp-01-dblc6195cda8)$ gcloud iam roles create viewer --project $DEVSHHELL_PROJECT_ID \
--title "Role Viewer" --description "Custom role description." \
--permissions compute.instances.get,compute.instances.list --stage ALPHA
Created role [viewer].
description: Custom role description.
etag: BwZTKF--SN4=
includedPermissions:
- compute.instances.get
- compute.instances.list
name: projects/qwiklabs-gcp-01-dblc6195cda8/roles/viewer
stage: ALPHA
title: Role Viewer
student_01_55c96268f1c6@cloudshell:~ (qwiklabs-gcp-01-dblc6195cda8)$
```

Task 5 - List Custom Roles

```
gcloud iam roles list --project $DEVSHHELL_PROJECT_ID
```

Task 6 - Update Custom Roles

Updated the editor role via YAML adding storage permissions, and the viewer role via flags.

```
gcloud iam roles update editor --project $DEVSHHELL_PROJECT_ID --file new-role-
definition.yaml
gcloud iam roles update viewer --project $DEVSHHELL_PROJECT_ID --add-permissions
storage.buckets.get,storage.buckets.list
```

Task 7 - Disable a Custom Role

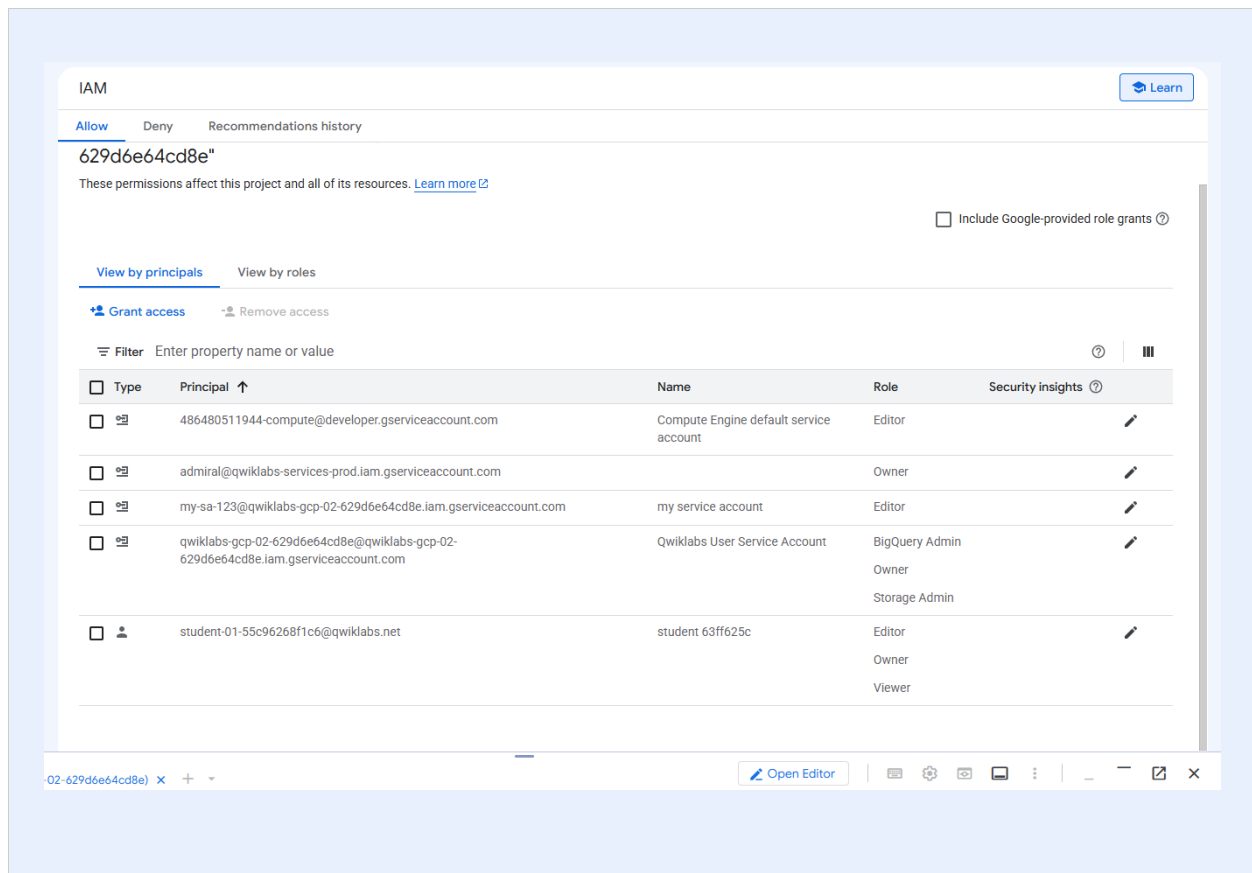
```
gcloud iam roles update viewer --project $DEVSHHELL_PROJECT_ID --stage DISABLED
```

Task 8 - Delete a Custom Role

```
gcloud iam roles delete viewer --project $DEVSHHELL_PROJECT_ID
```

Task 9 - Restore a Custom Role

```
gcloud iam roles undelete viewer --project $DEVSHHELL_PROJECT_ID
```



What I Learned

- Custom roles enforce least privilege by including only needed permissions
- Permissions are formatted as service.resource.verb (e.g. compute.instances.list)
- Custom roles can be created via YAML files or CLI flags
- Deleted roles can be restored within 7 days before permanent deletion
- Disabling a role inactivates all policy bindings without deleting the role

Resources

GCP Custom Roles Docs: <https://cloud.google.com/iam/docs/creating-custom-roles>

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